DS2002 Midterm Project | Fall 2026

# Walmart Hurricane Analytics

What Florida shoppers bought before Hurricane Helene (September 2024)

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The data

# The raw data had a lot of problems

### 60 store IDs for 12 stores

FL-239, FL239, FL_239 and fl-239 are all the same Orlando store. The store file also listed Orlando and Kissimmee twice.

### 4 SKUs for one Pop-Tart

Strawberry Pop-Tarts were split across PT-12, POPTART-STRAW, PT_STRAWBERRY and SKU#459812. Overall, 45 SKUs were really 20 products.

### 886 missing timestamps

Missing values represented 3.5% of the transactions and were written 5 different ways: NULL, N/A, NaN, nan and blank.

### Other issues

30 duplicate rows, 130 negative quantities, prices stored as text, and 37 category names for 10 categories.

The pipeline

# How we cleaned and joined the data

- Our cleaning function, clean_all(), runs 11 steps, and we logged 32 decisions
- Transactions went from 25,562 rows to 25,532 rows
- We mapped 45 SKUs down to 20 products
- We pulled weather data, including rain, max wind, and lowest pressure, from the Open-Meteo Historical API
- We used Eastern time for our weather data so it matched the sales dates.
    - If we used UTC, 8 of every 48 hourly readings would have matched the wrong day
- The join kept all 30 sales days, and the unit total did not change

## Decision 1: timestamps

We kept the 884 rows with no timestamp.

Dropping them would remove real sales, but if we filled in a date we would just be guessing. We used these rows in our totals but left them out of anything by date. (The raw file had 886, and 2 were duplicate rows we removed.)

## Decision 2: negative units

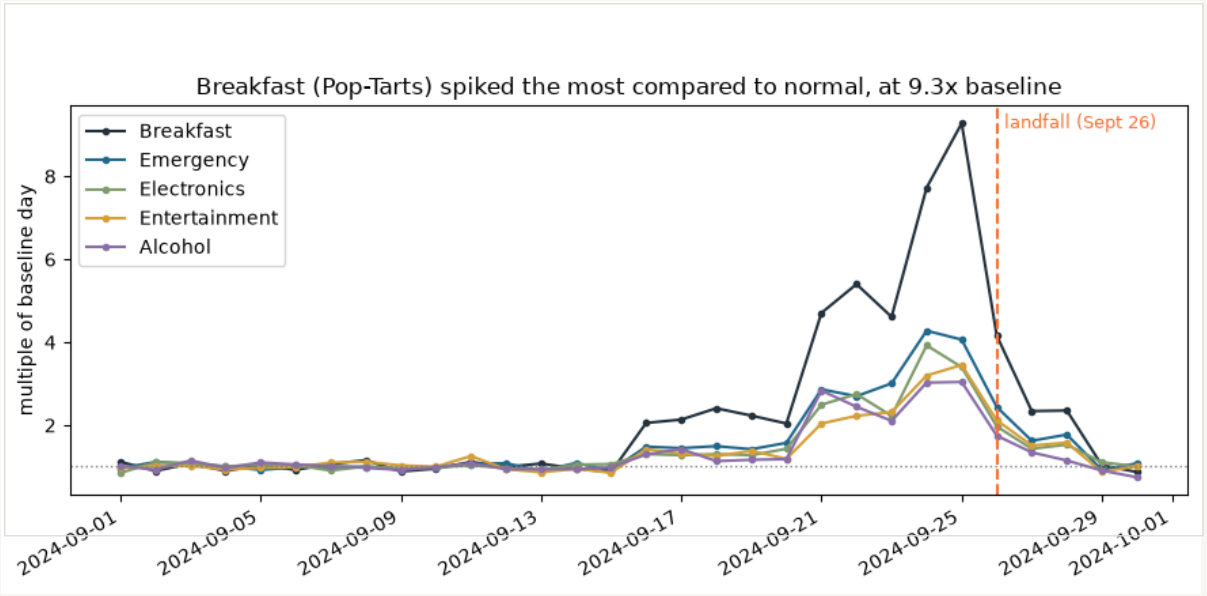
We flipped 130 negative quantities.

None matched a return, so we treated them as sign typos. Total units went from 49,343 to 49,825.

Q1 Demand Surge

# Breakfast (Pop-Tarts) had the biggest surge at +414%

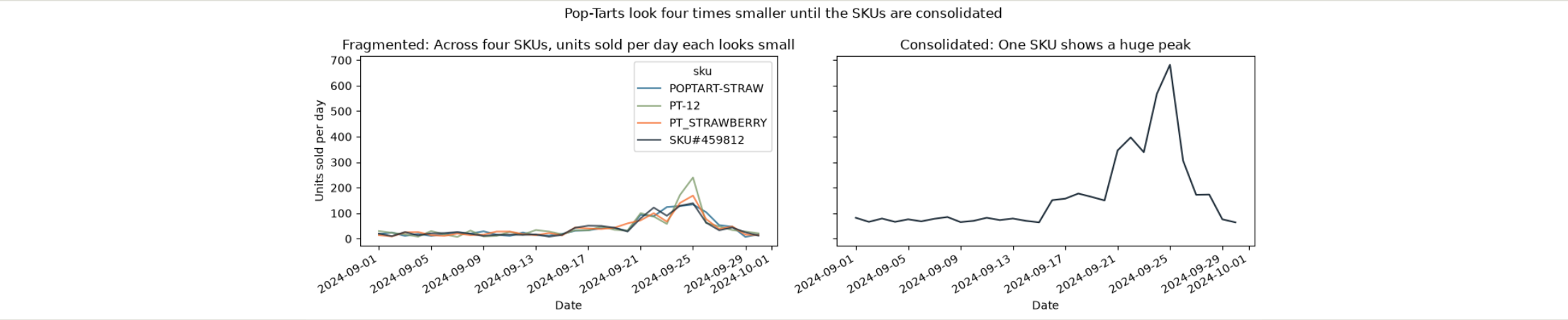
- Breakfast went from 73.5 to 378 units a day (5.1x, or +414%)
- Emergency +184%, Electronics +150%, Entertainment +126%, Alcohol +126%
- Snacks only went up 25%, so people were not buying more of everything
- Sales started rising Sept 16, peaked Sept 24 and 25, and dropped on landfall day




**Chosen time periods:**
Baseline: Sept 1 to 15. Surge: Sept 19 to 25. Landfall: Sept 26

Q2 The Pop-Tarts Effect

# Combining the 4 Pop-Tart SKUs changed the top product



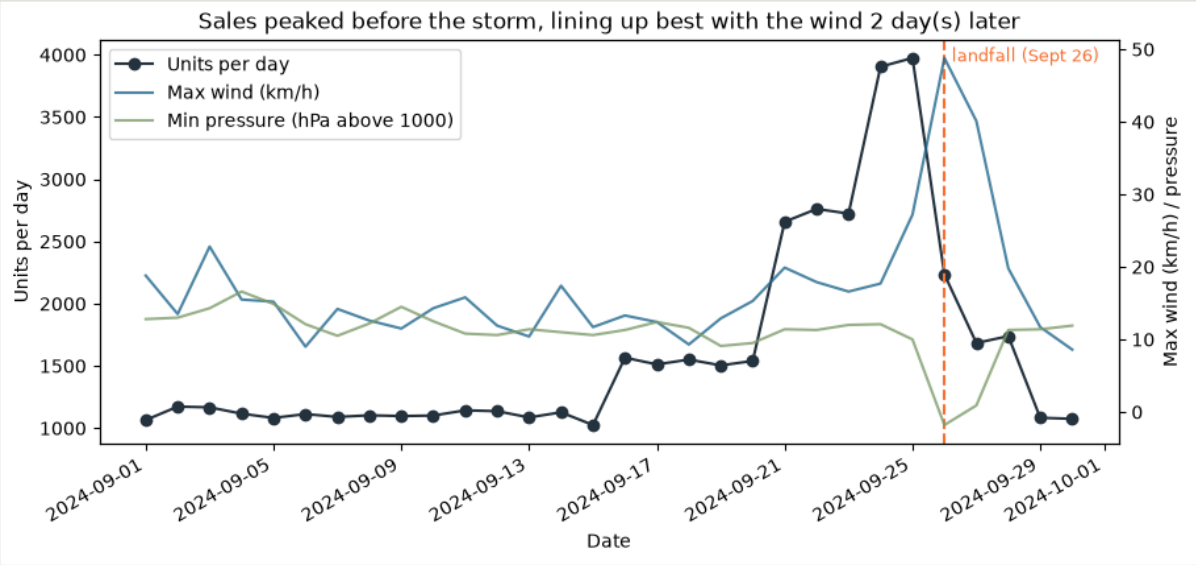

| | Before combining | After combining |
|---|---|---|
| Units per day in the surge | 103 (PT-12) | 378 |
| Rank in the surge | 4th, behind 3 water SKUs | 1st, ahead of water (321) |
| September total | 1,385 | 5,218 (3.8x more) |

- Peak: 682 units on Sept 25, 9.3x the baseline
- Without combining the SKUs, Walmart would not order enough of the top product

Q3 Weather Correlation

# People stocked up about 2 days before the worst wind

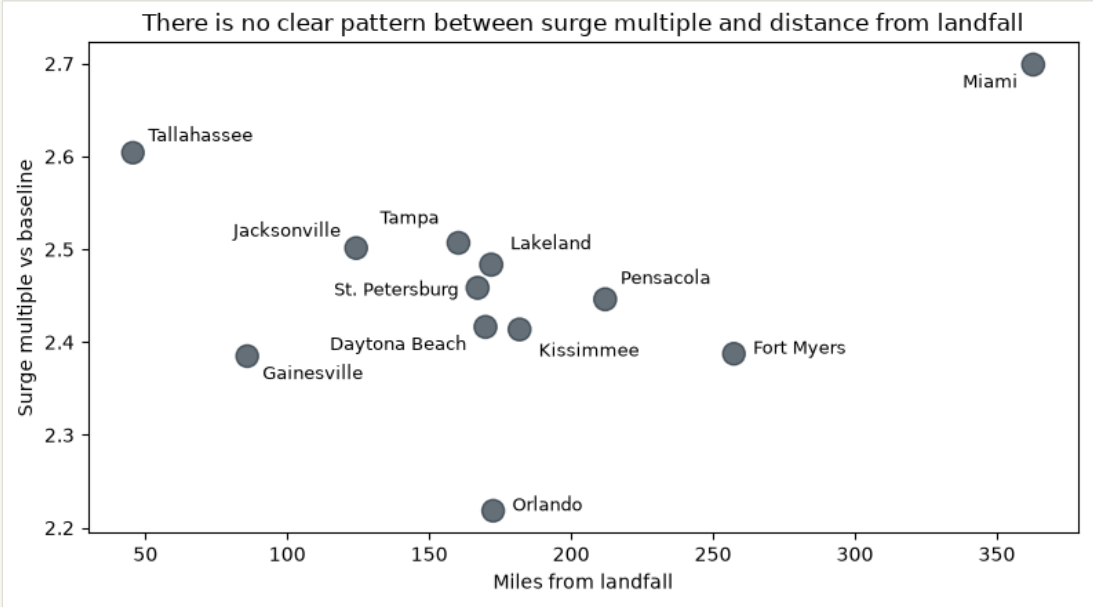
- Same day: wind r = 0.40, pressure r = -0.22, rain r = -0.26
- Weather 2 days later: wind r = 0.82, pressure r = -0.76
- Every category was strongest at 2 days (Health r = 0.85, Snacks r = 0.67)
- Rain stayed weak at every lag
- People stocked up about 2 days before the worst weather
- Note: weather is from Orlando only, and the 2-day correlations use 28 days



Q4 Stores and Geography

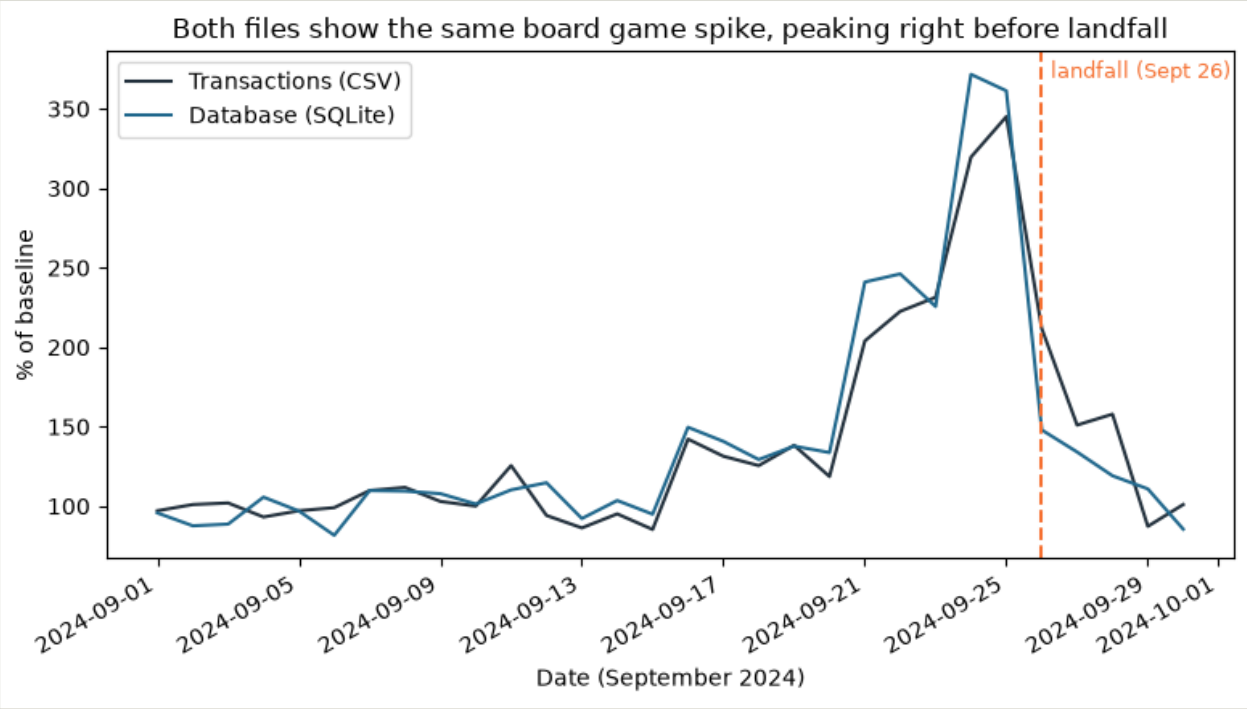
# Distance from landfall did not explain the surge

- All 12 stores surged between 2.2x and 2.7x their baseline
- Distance vs. surge: r = 0.23 across 12 stores (farther stores surged slightly more)
- Every store peaked on Sept 24 or 25
- Miami was the highest (2.70x) and Orlando was the lowest (2.22x)
- Miami's Pop-Tarts surged 7.2x vs. 4.0x to 5.8x at other stores, but on a baseline of only about 5 units a day



Q5 Board Games: Signal or Artifact?

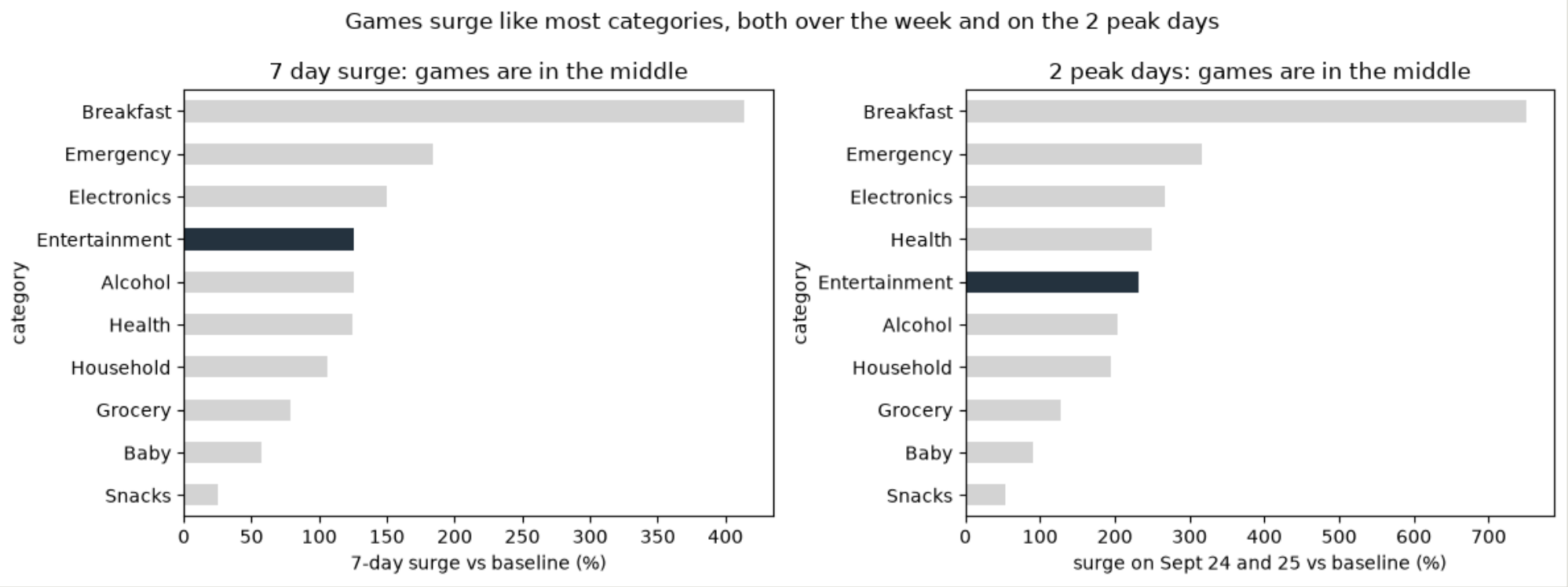
# The board game surge is real, but +250% is only the 2 peak days




- On Sept 24 and 25: +232% in the transactions and +267% in the database
- Over the full 7 days: +126% in the transactions and +145% in the database

Q5 Board Games: Signal or Artifact?

# Entertainment sat in the middle of the 10 categories




- Games surged like most categories, both over the 7 day surge and on the 2 peak days
  - Entertainment ranked 5$^{th}$ of 10 categories over the peak days, and 4$^{th}$ over the 7 days
- Breakfast, Emergency and Electronics rose more, and Snacks rose the least
- So it is a normal storm peak, not something special to games

Q5 Board Games: Signal or Artifact?

# We checked for artifacts and did not find any

| What we checked | What we found |
|---|---|
| Both data sources | The transactions file and the database move together day by day (r = 0.96) |
| Category lumping | Both products went up (Monopoly +164%, Playing Cards +98%) and all 12 stores surged |
| SKU miscoding | Monopoly SKUs all sell for about $20 and card SKUs for about $5. No game names outside Entertainment |
| Inventory reporting lag | Sales and stock both changed on Sept 21, so no lag |
| A few big orders | Transactions per day went from 57.9 to 133.3, with about the same units per purchase (1.8 vs. 1.7) |

Recommendation

# What Walmart should do before the next storm

1. Stock every Florida store, not just the ones closest to the storm path (Q4: every store surged 2.2x to 2.7x)
2. Order about 5x normal Pop-Tarts for the week before landfall, and up to 9x for the 2 days before (Q1 and Q2: 5.1x surge, 9.3x peak)
3. Give Emergency items the most truck space (1,011 units a day in the surge)
4. Have stores fully stocked 5 days before landfall, since sales jump then and peak 2 days before the worst wind (Q3)
5. Order about 2.6x normal Monopoly and 2.0x normal playing cards, but after water, batteries, flashlights and Pop-Tarts (Q5: Entertainment ranked 5th of 10 on the peak days)

Limits

# What our results cannot tell Walmart yet

- Weather comes from Orlando only, so we could not compare weather store by store
- We studied one storm, so the surge sizes and the 2 day lead may not repeat
- The 2 day correlations use only 28 days of data
- We had no cost or margin data, so we could not size the cost of overstocking
- **Next step:** repeat this on several storms and add cost data

# Thank you

Questions?